

Report
Of
Full Stack Development

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 25SCS1003000491
Name of Student: Aquib Ali Khan
Batch: 2024 - 2028

CANDIDATE’S DECLARATION

I, Aquib Ali Khan, do hereby declare that the internship report titled “**Cyber Security**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name : **Aquib Ali Khan**

Roll No. : **25SCS1003000491**

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout my internship at Thiranex. This internship gave me an opportunity to gain practical knowledge in cybersecurity and to apply my technical skills through project-based tasks.

I am thankful to the organization and the mentors for providing me with guidance, constructive feedback, and support during the internship. Their guidance helped me understand practical approaches to cybersecurity and improve my programming, problem-solving, and technical skills.

During the internship, I worked on different cybersecurity projects, including the Password Strength Analyzer, Vulnerability Scanner, Phishing Email Detection Model, and Secure Login System. Working on these projects helped me gain practical exposure to password security, vulnerability assessment, phishing detection, authentication, secure programming, testing, and basic cybersecurity practices.

I would also like to express my gratitude to my teachers and academic mentors for their continuous support and encouragement throughout the internship. Their guidance helped me maintain a systematic approach towards learning and completing the assigned project work.

Finally, I express my heartfelt gratitude to my parents and family for their constant encouragement, support, and blessings throughout this internship. Their motivation helped me stay focused and complete the internship work successfully.

Signature

Student Name : **Aquib Ali Khan**

Roll No. : **25SCS1003000491**

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3. Internship Completion Certificate



3.1 Internship Offer Letter



Date: 16 August 2026

TO,

Aquib Ali Khan

Intern ID: THX-AUG1626-174

Subject: Offer Letter for Internship in Cyber Security

Dear **Aquib Ali Khan**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - Cyber Security
Commencement Date	16 Aug 2026
Completion Date	15 Sept 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.
www.thiranex.in



4. Project Description

The internship provided practical exposure to cybersecurity through a series of project-based tasks. The projects were designed to develop an understanding of common security problems and their practical solutions. During the internship, the work progressed from basic password security and vulnerability identification to phishing detection and secure user authentication.

The four projects completed during the internship were:

1. Password Strength Analyzer
2. Vulnerability Scanner (Mini Project)
3. Phishing Email Detection Model
4. Secure Login System

These projects covered different areas of cybersecurity, including password security, vulnerability assessment, phishing detection, authentication, input validation, and secure application development. The project work also provided practical experience in analyzing security-related problems, implementing solutions, testing functionality, and understanding common security risks.

The projects were completed according to the requirements provided during the internship. Each project focused on a specific security-related problem and helped in developing practical knowledge through implementation and testing.

4.1 Introduction

Cybersecurity is an important area of computer science that focuses on protecting systems, applications, networks, and user information from unauthorized access and other security threats. With the increasing use of web applications and digital services, understanding common security issues has become an important part of software development.

The internship provided an opportunity to study and implement basic cybersecurity concepts through practical projects. Instead of focusing only on theoretical concepts, the assigned tasks required the development of working solutions for different security-related requirements.

The first project was the Password Strength Analyzer, which focused on evaluating user-entered passwords based on factors such as length, complexity, and uniqueness. The project also involved providing suggestions for creating stronger password alternatives.

The second project was the Vulnerability Scanner (Mini Project). It focused on identifying common vulnerabilities by checking areas such as open ports, weak configurations, and outdated software versions. The project also involved generating a simple vulnerability report based on the findings.

The third project was the Phishing Email Detection Model. This project involved developing a machine learning model using Scikit-learn to classify emails as either “Phishing” or “Safe”. The model used email-related features such as URLs, keywords, and other textual characteristics and included evaluation through accuracy and a confusion matrix.

The fourth project was the Secure Login System. This project focused on developing a secure login web application with user registration, authentication, hashed passwords, input validation, session management, and protection against common attacks such as SQL injection. Optional two-factor authentication was also considered as part of the project requirements.

Together, these projects provided practical exposure to different areas of cybersecurity. They helped in understanding how security concepts can be applied while developing software and how common security risks can be identified, analyzed, and reduced through appropriate techniques.

The internship also helped in improving programming, logical reasoning, testing, debugging, and problem-solving skills. Working on different security-oriented projects provided a better understanding of the importance of secure development practices in modern applications.

4.2 Organization Profile

The internship was undertaken at Thiranex as part of a practical training program focused on cybersecurity and technical skill development. The internship provided an opportunity to work on project-based tasks and understand how cybersecurity concepts can be applied while developing and testing software applications.

The training approach focused on practical implementation rather than limiting the learning process to theoretical concepts. During the internship, different security-related projects were assigned to develop an understanding of password security, vulnerability identification, phishing detection, and secure authentication.

The internship involved working on multiple projects with different requirements and levels of implementation. These projects provided exposure to areas such as password analysis, vulnerability scanning, machine learning-based classification, secure authentication, input validation, password hashing, session management, and security testing.

As part of the internship, four projects were completed: Password Strength Analyzer, Vulnerability Scanner (Mini Project), Phishing Email Detection Model, and Secure Login System. Each project addressed a different cybersecurity requirement and provided practical experience in developing security-oriented solutions.

The internship also encouraged the use of structured development practices such as understanding requirements, planning the solution, implementing the required features, testing the application, identifying errors, and making necessary improvements.

Overall, the internship served as a practical learning environment for developing cybersecurity knowledge and improving programming and problem-solving abilities. It helped in understanding common security risks and the importance of applying appropriate security measures during software development.

4.3 Problem Statement

The increasing use of digital systems and web applications has also increased the need for proper security measures. Weak passwords, exposed services, outdated software, phishing messages, and insecure authentication mechanisms can create security risks for users and applications. Therefore, the internship projects were designed to address different common cybersecurity problems through practical implementation.

The first problem was related to weak and easily guessable passwords. Users often create passwords that are short, simple, or based on common patterns. Such passwords can increase the possibility of unauthorized access. The Password Strength Analyzer was developed to examine password characteristics and provide feedback for improving password strength.

The second problem involved identifying basic vulnerabilities in a system. Open ports, weak configurations, and outdated software can provide opportunities for security attacks. The Vulnerability Scanner project was developed to check these areas and provide information about potential vulnerabilities.

The third problem was related to phishing emails. Phishing messages may appear to be legitimate but can contain suspicious links, keywords, or other characteristics intended to mislead users. The Phishing Email Detection Model was developed to classify emails into “Phishing” and “Safe” categories using relevant email features and machine learning techniques.

The fourth problem concerned insecure login systems. Poor password handling, insufficient input validation, and weak session management can expose user accounts to security threats. The Secure Login System was developed to demonstrate safer user registration and authentication practices, including password hashing, input validation, session management, and protection against common attacks.

A common challenge across these projects was understanding how different security threats can affect software systems and how appropriate security techniques can be applied to reduce these risks. Testing and debugging were also required to verify that the implemented solutions worked according to their intended requirements.

Thus, the overall problem addressed during the internship was to develop practical cybersecurity solutions for common security concerns while gaining hands-on experience in secure application development, vulnerability identification, threat detection, and authentication.

4.4 Project Objectives

The main objective of the internship was to develop practical knowledge of cybersecurity by working on different security-related projects. The internship focused on understanding common security problems and implementing suitable solutions through hands-on project work.

The major objectives of the project work were:

1. To understand the basic concepts and practical applications of cybersecurity.
2. To develop a Password Strength Analyzer for evaluating password strength and identifying weak password characteristics.
3. To understand the factors that make passwords strong, including password length, complexity, and uniqueness.
4. To develop a Vulnerability Scanner capable of checking for common security issues such as open ports, weak configurations, and outdated software versions.
5. To understand the basic process of vulnerability identification and security assessment.
6. To develop a Phishing Email Detection Model for distinguishing between phishing and safe emails.
7. To apply machine learning techniques using Scikit-learn for phishing email classification.
8. To evaluate the phishing detection model using suitable measures such as accuracy and a confusion matrix.
9. To develop a Secure Login System with user registration and authentication functionality.
10. To implement password hashing so that user passwords are not stored directly in their original form.
11. To implement input validation and session management as part of secure authentication.
12. To understand and apply basic protection against common attacks such as SQL injection.
13. To improve practical skills in programming, testing, debugging, and problem-solving through cybersecurity projects.
14. To understand the importance of secure coding practices while developing software applications.
15. To gain practical experience in analyzing security-related problems and developing working solutions based on defined requirements.

4.5 Scope of the Project

The scope of the internship covers the development of four cybersecurity-based projects, with each project focusing on a different security requirement. Together, the projects provide practical exposure to password security, vulnerability assessment, phishing detection, and secure authentication.

The Password Strength Analyzer focuses on evaluating the security level of passwords entered by users. Its scope includes checking password length, complexity, and uniqueness and providing suggestions that can help users create stronger passwords.

The Vulnerability Scanner (Mini Project) focuses on identifying common security weaknesses in a system. Its scope includes checking for open ports, weak configurations, and outdated software versions. The project also provides a basic vulnerability report based on the identified issues.

The Phishing Email Detection Model focuses on detecting potentially harmful emails. Its scope includes preparing email-related features, applying a machine learning model using Scikit-learn, and classifying emails into “Phishing” or “Safe” categories. The model can also be evaluated using accuracy and a confusion matrix.

The Secure Login System focuses on implementing a safer user authentication process. Its scope includes user registration, login authentication, password hashing, input validation, and session management. The project also considers protection against common attacks such as SQL injection, with two-factor authentication included as an optional security feature.

From a technical perspective, the overall scope includes security analysis, application development, machine learning-based classification, authentication, data validation, password protection, vulnerability identification, testing, and debugging.

The projects also provide a foundation for future improvements. Additional security checks, improved vulnerability detection, more advanced phishing detection techniques, stronger authentication mechanisms, improved reporting, and additional security controls can be incorporated depending on future requirements.

4.6 Technologies and Tools Used

The development of the internship projects involved the use of different programming languages, libraries, frameworks, databases, development tools, and security-related technologies. The technologies were selected according to the requirements of each project and the type of security problem being addressed.

4.6.1 Frontend Technologies

HTML was used to create the basic structure of web pages and application interfaces. It provided the elements required for displaying forms, information, and other components of the web applications.

CSS was used for styling the interfaces and improving their visual presentation. It was used to manage layouts, spacing, fonts, and responsive behaviour of the web-based projects.

JavaScript was used to implement interactive functionality and client-side behaviour. It helped in handling user inputs, performing validations, and connecting frontend components with application logic.

Python was used for cybersecurity-related development and machine learning tasks. It provided the programming environment required for implementing the Password Strength Analyzer, Vulnerability Scanner, and Phishing Email Detection Model according to their respective requirements.

4.6.2 Backend Technologies

Scikit-learn was used for developing the Phishing Email Detection Model. It provided the machine learning tools required for training and evaluating the classification model used to identify phishing and safe emails.

Password analysis techniques were used in the Password Strength Analyzer to examine characteristics such as password length, complexity, and uniqueness.

Network and vulnerability scanning techniques were used in the Vulnerability Scanner to identify areas such as open ports, weak configurations, and outdated software versions.

Password hashing was used in the Secure Login System to protect user passwords before storing them. This helped avoid storing passwords directly in their original form.

4.6.3 Database

Backend technologies were used where server-side processing and authentication functionality were required. The backend handled application requests, user-related operations, validation, and communication with the database.

A database was used for storing information required by the Secure Login System, including user-related data and authentication information. Appropriate database operations were used to create, retrieve, and manage application records.

4.6.4 Authentication and Security

Authentication mechanisms were implemented in the Secure Login System to verify users before allowing access to protected functionality.

Input validation was used to check user-provided information and reduce the possibility of invalid or unsafe input being processed by the application.

Session management was used to maintain authenticated user sessions after successful login.

SQL injection protection was considered as an important security requirement for preventing malicious database queries through user input.

4.6.5 Development and Version Control Tools

Visual Studio Code was used as the primary development environment for writing and managing project source code.

Git was used for version control and for maintaining changes made during the development process.

GitHub was used for managing project repositories and maintaining the source code remotely.

Postman was used where required for testing backend APIs and checking request-response behaviour.

4.6.6 Supporting Technologies

Different supporting libraries, development tools, and testing techniques were used according to the requirements of individual projects. These technologies helped in implementing application functionality, testing security-related features, identifying errors, and improving the overall reliability of the projects.

The combination of these technologies provided a practical development environment for working on the internship projects. Programming technologies were used for implementation, security techniques were applied to address common risks, machine learning tools supported phishing detection, and development and testing tools helped in managing and verifying the project work.

4.7 System Architecture

The system architecture of the internship work consists of four cybersecurity projects developed to address different security-related requirements. Although the purpose and implementation of each project are different, they follow a structured approach in which user input, application logic, security processing, and output work together.

The overall architecture can be understood through the major components involved in the individual projects, including the user interface, application logic, security processing, database, and machine learning components where required.

4.7.1 User Interface Layer

The user interface layer represents the part of the application through which users interact with the projects. It is responsible for accepting user input, displaying results, and providing information about the security analysis performed by the application.

HTML, CSS, JavaScript, and other suitable frontend technologies were used according to the requirements of the individual projects. For example, the Password Strength Analyzer accepts a password from the user and displays its strength and improvement suggestions.

Similarly, the Secure Login System provides interfaces for registration and login, while the Phishing Email Detection Model allows email-related information to be provided for classification.

4.7.2 Application and Security Processing Layer

The application and security processing layer contains the main logic of the projects. It processes the input received from the user and applies the required security techniques.

In the Password Strength Analyzer, the input password is examined according to defined strength criteria. In the Vulnerability Scanner, the application performs the required checks to identify security weaknesses such as open ports, weak configurations, and outdated software versions.

For the Phishing Email Detection Model, the input email is processed and passed through the trained machine learning model to determine whether it belongs to the “Phishing” or “Safe” category.

The Secure Login System uses application logic for registration, authentication, password hashing, input validation, and session management.

4.7.3 Database Layer

The database layer is used where persistent application data is required. In the Secure Login System, user-related information is stored and managed through the database.

The application communicates with the database through the backend so that user records and authentication-related information can be created, retrieved, and managed securely.

The use of password hashing ensures that passwords are not stored directly in their original form.

4.7.4 Machine Learning Component

The Phishing Email Detection Model includes a separate machine learning component. Email-related features are processed and provided to a classification model developed using Scikit-learn.

The trained model analyzes the available features and produces a classification result indicating whether an email is “Phishing” or “Safe”. The performance of the model can be evaluated using accuracy and a confusion matrix.

4.7.5 Project-Level Architecture

The architecture of the four projects can be summarized as follows:

Password Strength Analyzer:

User → Password Input → Strength Analysis → Strength Result and Suggestions

Vulnerability Scanner:

User → Scanner Input → Security Checks → Vulnerability Analysis → Vulnerability Report

Phishing Email Detection Model:

Email Input → Feature Processing → Machine Learning Model → Classification → Phishing / Safe

Secure Login System:

User → Registration/Login → Backend → Authentication & Validation → Database

↓

Password Hashing / Session Management

The architecture provides a separation between user interaction, application processing, security mechanisms, and data management. This structure makes the projects easier to understand, test, maintain, and improve. It also allows individual security components to be modified or enhanced without requiring major changes to the complete application.

4.8 Methodology

The projects developed during the internship followed a structured and practical development approach. The work started with understanding the requirements of each project and then planning, implementing, testing, and improving the solution. Since the projects were based on different

cybersecurity requirements, the development process was adjusted according to the purpose of each project.

4.8.1 Requirement Analysis

The first stage involved understanding the requirements of each assigned project. The main security problem, required features, expected inputs, and expected outputs were identified before starting the implementation.

For the Password Strength Analyzer, the requirements related to password length, complexity, uniqueness, and suggestions were considered. Similarly, the requirements for vulnerability scanning, phishing detection, and secure authentication were studied before development.

4.8.2 System Planning and Design

After understanding the requirements, the basic structure of each project was planned. The required components, application flow, input handling, processing methods, and output were identified.

The planning stage helped in deciding how the user would interact with the application and how the security-related processing would be performed.

4.8.3 Application Development

The applications were developed according to their individual requirements. Programming languages and supporting technologies were selected based on the type of project.

The Password Strength Analyzer was developed to analyze password characteristics and provide strength-related feedback. The Vulnerability Scanner was implemented to perform security checks and identify potential vulnerabilities.

The Phishing Email Detection Model involved preparing relevant email features and applying a machine learning approach for classification. The Secure Login System was developed with registration and login functionality along with security mechanisms such as password hashing and input validation.

4.8.4 Security Feature Implementation

Security features were implemented according to the requirements of the individual projects. Password analysis techniques were used to identify weak passwords, while vulnerability scanning techniques were applied to identify common system weaknesses.

For the phishing detection project, machine learning techniques were used to classify emails. In the Secure Login System, password hashing, input validation, session management, and protection against SQL injection were considered to improve the security of the authentication process.

4.8.5 Data and Database Integration

Projects requiring data storage were connected with a database for maintaining application information. The Secure Login System used database operations for managing user-related records.

The application handled the required operations through the backend rather than allowing direct access to stored data from the user interface.

4.8.6 Machine Learning Implementation

The Phishing Email Detection Model was developed using Scikit-learn. Relevant features from email data were used for training and classification.

After training the model, its performance was evaluated using suitable evaluation methods, including accuracy and a confusion matrix. The model was then used to classify emails as either “Phishing” or “Safe”.

4.8.7 Testing and Debugging

After implementing the required features, the projects were tested to identify errors and unexpected behaviour. Different inputs and application conditions were checked to verify whether the implemented functionality produced the expected results.

Errors found during testing were analyzed and corrected through debugging. Testing was repeated after making changes to ensure that the modifications did not affect other working features.

4.8.8 Security Testing

Security-related functionality was also checked during development. Password analysis was tested using passwords with different levels of strength, while vulnerability scanning was checked against the required security conditions.

The phishing detection model was evaluated using its classification results. The Secure Login System was tested for authentication, input validation, password handling, session behaviour, and protection against common attacks such as SQL injection.

4.8.9 Version Control and Project Management

Git was used to maintain changes made during project development, while GitHub was used for remote repository management. The source code and project files were maintained in an organized manner so that development changes could be tracked.

4.8.10 Final Review

The final stage involved reviewing the completed projects and checking whether the implemented features satisfied their intended requirements. The applications were reviewed for functionality, security-related behaviour, usability, and proper handling of inputs and outputs.

This methodology provided a systematic approach to completing the internship projects and helped in understanding the practical process of developing, testing, and improving cybersecurity-based applications.

4.9 Expected Outcomes

The internship was expected to provide practical knowledge of cybersecurity and improve the ability to apply security concepts while developing software applications. The completion of the four projects was expected to provide experience with different types of cybersecurity requirements and their practical implementation.

The major expected outcomes of the internship were:

1. Development of a functional Password Strength Analyzer capable of evaluating password strength and providing suggestions for creating stronger passwords.
2. Practical understanding of password security factors such as length, complexity, and uniqueness.
3. Development of a Vulnerability Scanner (Mini Project) for identifying common security issues such as open ports, weak configurations, and outdated software versions.
4. Practical understanding of the basic process of vulnerability identification and security assessment.
5. Development of a Phishing Email Detection Model capable of classifying emails as “Phishing” or “Safe”.
6. Practical experience in using Scikit-learn and machine learning techniques for email classification.
7. Evaluation of the phishing detection model using accuracy and a confusion matrix to understand its classification performance.
8. Development of a Secure Login System with user registration and authentication functionality.

9. Practical understanding of password hashing, input validation, and session management for improving login security.
10. Understanding of basic protection against common security threats such as SQL injection.
11. Improved ability to identify, analyze, and resolve errors through regular testing and debugging.
12. Better understanding of secure coding practices and the importance of security during application development.
13. Practical experience in working with different cybersecurity tools, programming technologies, and development environments.
14. Improvement in programming, logical reasoning, problem-solving, and software development skills through hands-on project work.
15. Development of a basic understanding of how cybersecurity techniques can be applied to protect applications and user information.

Overall, the expected outcome of the internship was to develop a stronger practical understanding of cybersecurity and gain experience in converting security-related requirements into functional software solutions. The projects were also expected to provide a foundation for further learning in cybersecurity, secure software development, and application security.

4.10 Certificates of Completion and Other Communication Mail Proof

The internship was undertaken at Thiranex as part of a practical training program focused on cybersecurity and project-based learning. The internship provided an opportunity to work on different cybersecurity projects and apply technical concepts through practical implementation.

The internship involved working on four major projects: Password Strength Analyzer, Vulnerability Scanner (Mini Project), Phishing Email Detection Model, and Secure Login System. These projects provided practical exposure to different areas of cybersecurity, including password security, vulnerability assessment, phishing detection, machine learning, authentication, password hashing, input validation, and session management.

The internship offer letter and other official communication received during the internship served as supporting evidence of the internship engagement and assigned work. The communication records also provided information related to the internship role, project requirements, instructions, and submission activities.

The following documents are attached in this section:

1. Internship Offer Letter issued by Thiranex.
2. Project assignment and requirement records.
3. Project submission records.
4. Relevant internship communication and instructions.

THIRANEX VIRTUAL INTERNSHIPS

Official Certificate Ready for Issuance

Dear **Aquib Ali Khan**,

Your internship task submissions for **Cyber Security** have been evaluated and **APPROVED**.

Intern ID:	THX-AUG1626-174
Domain:	Cyber Security
Evaluation Status:	✓ Verified & Approved

Your official MSME-registered Internship Completion Certificate is ready for verification. Complete the process below to claim your credential.

[Claim & Download Certificate →](#)

5. Internship Completion Certificate.

These documents provide supporting evidence of the internship, the assigned project work, project submissions, and completion of the internship requirements.

5. Bibliography / References

1. Thiranex, Internship Offer Letter for Internship in Cybersecurity, 2026.
2. Mozilla Developer Network (MDN), Web Development Documentation, documentation related to HTML, CSS, JavaScript, HTTP methods, and web application development.
3. Python Software Foundation, Python Documentation, documentation related to Python programming and application development.
4. Scikit-learn, Scikit-learn Documentation, documentation related to machine learning algorithms, classification, model evaluation, and confusion matrix.
5. OWASP, OWASP Web Security Documentation, resources related to web application security, authentication, input validation, and common web vulnerabilities.
6. OWASP Foundation, OWASP SQL Injection Prevention Cheat Sheet, documentation related to preventing SQL injection attacks.
7. Git, Git Documentation, documentation related to distributed version control and source-code management.
8. GitHub, GitHub Documentation, documentation related to repositories, source-code management, and version-control workflows.
9. Python Standard Library Documentation, resources related to Python libraries and modules used during project development.
10. Relevant internship-provided project instructions, task requirements, and communication records used during the development of the assigned cybersecurity projects.